

Short-term Drought Impacts are Influenced by Annual Flow Patterns and Fish Life History Strategies in North Texas

Cassandra Rendon^a,

Seth Callahan^{ab}

Bjorn Schmidt^{a*}

^a Department of Biological and Environmental Science, East Texas A&M University, Commerce, Texas 75428, USA, rendon.cassandra@yahoo.com, Bjorn.Schmidt@tamuc.edu - corresponding author

^b current affiliation: Missouri Cooperative Fish and Wildlife Research Unit, University of Missouri, Columbia, Missouri 65211, USA, bscnh9@missouri.edu

OrCID:

Bjorn Schmidt: 0000-0002-9030-4037

Seth Callahan: 0009-0002-5228-9136

Abstract: Sixteen fish communities were assessed in North Texas, USA, in spring 2022 and 2023, separated in time by a severe drought. To assess the influence of flow patterns on drought response, mean low flow patterns in August and annual flow variation (flow stability) were measured for each site. Sites were then assigned drought risk scores based on these variables, and the data was split into higher drought risk and lower drought risk groupings. A Non-metric multidimensional scaling (NMDS) analysis of pre and post drought communities was performed, and paired Euclidean distance was measured to indicate annual change. Analyses showed that annual community change patterns were significantly stronger in the higher risk sites than the lower risk sites, and this pattern was mainly driven by species losses. To determine drought impacts relative to species life history strategies a dataset of six life history trait values for each species was used in another NMDS analysis together with Archetype Analysis to define more opportunistic, more equilibrium, and more periodic strategies among the fish species of the region. More opportunistic species increased in mean relative abundance after drought by 4.8%, while less opportunistic species (equilibrium and periodic) decreased by 2.7%, which indicated a significantly different annual response to the drought. This research shows that opportunistic vs non-opportunistic life history strategies can relate to species short term drought risk and response, and that annual low flow and variability patterns are good indicators of potential short-term community change following droughts, which can be useful to better understand species risk and community response in drought prone regions.

Keywords

Drought, Opportunistic, Disturbance, Short-Term Response, Life History

INTRODUCTION

Recent analysis and projections show that the world temperatures have increased in recent decades along with a decrease in relative humidity, leading to a prediction of a more arid future with an increase in occurrence and severity of drought (Sherwood and Fu, 2014). These predictions are supported by global analyses that show that the magnitude of drought events have increased since 2000 or 2010, based on different models (Vicente-Serrano et al., 2022). This pattern may be especially important for river systems that are in semi-arid regions with high diversity, which may be expected to have changes in diversity or distributions, based on increased drought effects (Gill et al., 2020). Therefore, a better understanding of specific risk factors for drought effects across sites and species is needed to better predict or mitigate community changes.

Droughts can significantly affect fish habitat structure including reduced stream flow, increased temperature, reduced water oxygen concentration, increased competition or predation, and increased exposure to diseases (Lennox et al., 2019). These environmental stressors may also affect the fish's ability to migrate, reproduce, and disperse (Keaton et al., 2005, Jones and Petreman, 2013). Migration for species who tend to reside in small streams may prove to be especially challenging under drought conditions, because entire segments of small or shallow streams might lose flow, leaving only small, isolated pools as refugia (Lennox et al., 2019). This can lead to population fragmentation and bottleneck effects (Magoulick and Kobza, 2003). However, larger streams are usually less affected by extreme droughts and tend to maintain water flow through drought. Therefore, many fish might try to relocate to larger, perennial streams before the habitat connections are lost (Lennox et al., 2019). This often makes larger downstream locations more abundant and diverse during drought conditions than smaller streams (Keaton et al., 2005). Other fish species may have a strategy to remain in refugia or quickly colonize streams after water flow returns, taking advantage of reduced competition in newly wetted habitats (Archdeacon et al., 2022).

During droughts, habitat sizes are reduced (especially in small streams) and only the most tolerant species survive by either finding refuge in isolated pools or traveling downstream to larger, but more crowded, habitats (Magoulick and Kobza, 2003). Seasonal droughts may have little to no effect on the assemblages of drought resistant species who have become adapted to regularly occurring dry periods (Matthews and Matthews, 2003). However, severe droughts, due to extended dry periods and increased incidence of stream drying, may lead to the elimination of drought sensitive stream species, and have stronger regional effects, including distributions of more drought tolerant species (Dodds et al., 2004, Matthews and Marsh-Matthews, 2003, Hughes et al., 2005, Hitt et al., 2020). Recovery after a drought is greatly dependent upon which species are able to recolonize and reproduce faster, thereby establishing self-sustaining local populations. Therefore, a species' response to drought and subsequent recovery/recolonization is influenced mainly by their life history traits (Dodds et al., 2004, Stanley et al., 2012, Hitt et al., 2020). Over long temporal periods, communities may match a loose equilibrium, whereby even though there may be short-term changes caused by disturbance, including drought, communities tend to fluctuate around an average over much longer time scales (Matthews and Marsh-Matthews, 2016). However, if droughts are severe enough, have high frequency intervals, or have other stressors in the system such as reservoirs or barriers to longitudinal movement, then regional species loss could occur, causing directional change in communities (Marshall et al., 2021).

Life history traits can be measured and used to classify fish taxa into three different strategies: equilibrium, opportunistic, and periodic. Equilibrium fish species tend to be of larger body size, live longer, reach maturity later, produce less eggs, and provide more parental care to those eggs (Winemiller and Rose, 1992). These species have life history traits similar to those of K-selected species (based on the r/K selection theory) (Hitt et al., 2020). In contrast, opportunistic species tend to be smaller in size, live shorter lives, reach maturity faster, produce more eggs, and provide less parental care than equilibrium species (Winemiller and Rose, 1992). Therefore, these species are similar in life history traits to those of r-selected species (Hitt et al., 2020). Lastly, periodic strategists are typically characterized by large body sizes, high fecundity, and their unique adaptability to habitats with seasonal variability in high flow and low flow water conditions (Winemiller and Rose, 1992). One of the ways in which they often adjust to these periodic high flow events include frequent movement between stream habitats.

Life history traits of fish species may have a significant effect on the recolonization patterns of stream species (Adams and Warren, 2005; Hitt et al., 2020). Typically, opportunistic species are quicker to recolonize ecologically disturbed areas because they are able to resist drought effects by finding refuge in deep residual pools and/or colonize and respawn rapidly to increase their repopulation rates once the stream flow returns. Equilibrium species, however, are usually slower to recolonize streams because they tend to migrate downstream at the onset of drought (if habitat connections persist) and then move back upstream to reproduce after the return of stream flow (Smith, 1982, Lennox et al., 2019, Hitt et al., 2020). Lastly, periodic species

recolonization patterns may be similar to those of opportunistic species because their high mobility within river networks and their adaptability to variable water conditions allows them to find refuge easier and recolonize faster once stream flow returns (Magoulick et al., 2021). Long-term datasets suggest that opportunistic species can dominate communities after multi-year droughts (Malone et al., 2022), but other studies suggest that some opportunistic species can decrease following severe drought depending upon other patterns such as reproductive guild and presence of egg predators (Perkin et al., 2015; Nguyen et al., 2022). Examining short term response to drought in different habitats and communities can provide guidance for characteristics that provide resistance or resilience to drought and help better understand community successional patterns after the disturbance (Lennox et al., 2019). Particularly, more studies are needed for short-term impacts and response to severe drought, which have increased intensity or duration. Examining short-term responses to specific drought events in relation to life history traits can be useful for predicting future community responses to flow alterations, which is relevant as droughts may be expected to increase in arid regions of the world in the future (Wheeler et al., 2018; Lennox et al., 2019; Malone et al., 2022).

For this project, we had completed standardized sampling of fish communities in tributaries in North Texas during the spring of 2022, which was subsequently followed in Summer and Fall of that year by a severe regional drought, which was the 13th driest year in Texas in last 128 years, classified as a severe drought (NOAA, 2022). The goal of this research was to resample streams in the same annual time period as pre-drought samples (Spring 2023), specifically after water had returned to dry streams in late winter, early spring of that year to comparatively examine patterns of change in relation to stream flow patterns and species life history traits. We hypothesized that community patterns would have greater change in streams that had lower annual flows and higher annual variability in flow (less groundwater dependent for flows), and that opportunistic species would exhibit greater resistance or resilience to drought compared with non-opportunistic species. This would provide an opportunity to examine the short term, annual response to severe drought, facilitated by the existence of pre-drought datasets in the region.

METHODS

Study Area and Site Selection

The Red River is a major southern river that starts in the Panhandle of Texas and runs along the border of Texas and Oklahoma before passing through Arkansas and feeding into the Atchafalaya River and eventually Mississippi River. The locations used for this study mainly include tributaries of the middle Red River in Texas, below Lake Texoma (mostly within the boundaries of Fannin County and Grayson County) (Fig. 1). In spring of 2022, 31 stream communities in tributaries of the Red River from Lake Texoma to Texarkana were sampled by students of East Texas A&M. After this sampling event, there was a severe drought that occurred in the region in summer/fall of 2022. In this study, we resampled 16 of these stream sites in spring 2023, after water returned to any dry streams to compare short-term severe drought impacts on fish communities. For this goal, 16 of the original 31 locations were available for resampling based on the following criteria – 1) sites were confined to the same region along the middle Red River to ensure a similar regional species pool in the analysis, 2) balance in the sites based on stream size to allow for assessing comparative flow patterns in the data, with eight smaller streams (cumulative upstream drainage areas less than 20 km²; average = 10.02 km²) and eight larger streams (cumulative upstream drainage areas less than 20 km²; average = 259.97 km²).

Fish Sampling

In order to maintain consistency in data collection, we followed the same methods used at each site in resamples as the spring 2022 fish sampling. Sampling procedures followed a modified fish collection procedure provided by the Texas Commission on Environmental Quality (TCEQ, 2014), which was modified for smaller

streams and small sampling crew. At each stream site a standardized 150m reach was established and fish collection was performed through the usage of a single pass backpack electro-shocker (Aquashock Solutions AS2), targeting all available habitat along both banks and in the main channel. Supplemental seine nets (6m net with 0.038m mesh) were used at sites that had pools with depths greater than 1.2 m, which precluded use of the electroshocker in these habitats. For each site with supplemental seine collection 3 seine pulls were done. Sites with supplemental seining in 2022 were the only ones that had supplemental seining in 2023, ensuring duplication of sampling methods. Fish were held in collection buckets, and identified in the field based on distinguishing characteristics or fixed in 10% formalin and identified with dichotomous keys in the lab (Craig and Bonner, 2019). For common and easily identifiable fishes identified in the field, photovouchers were used for each unique species at a site.

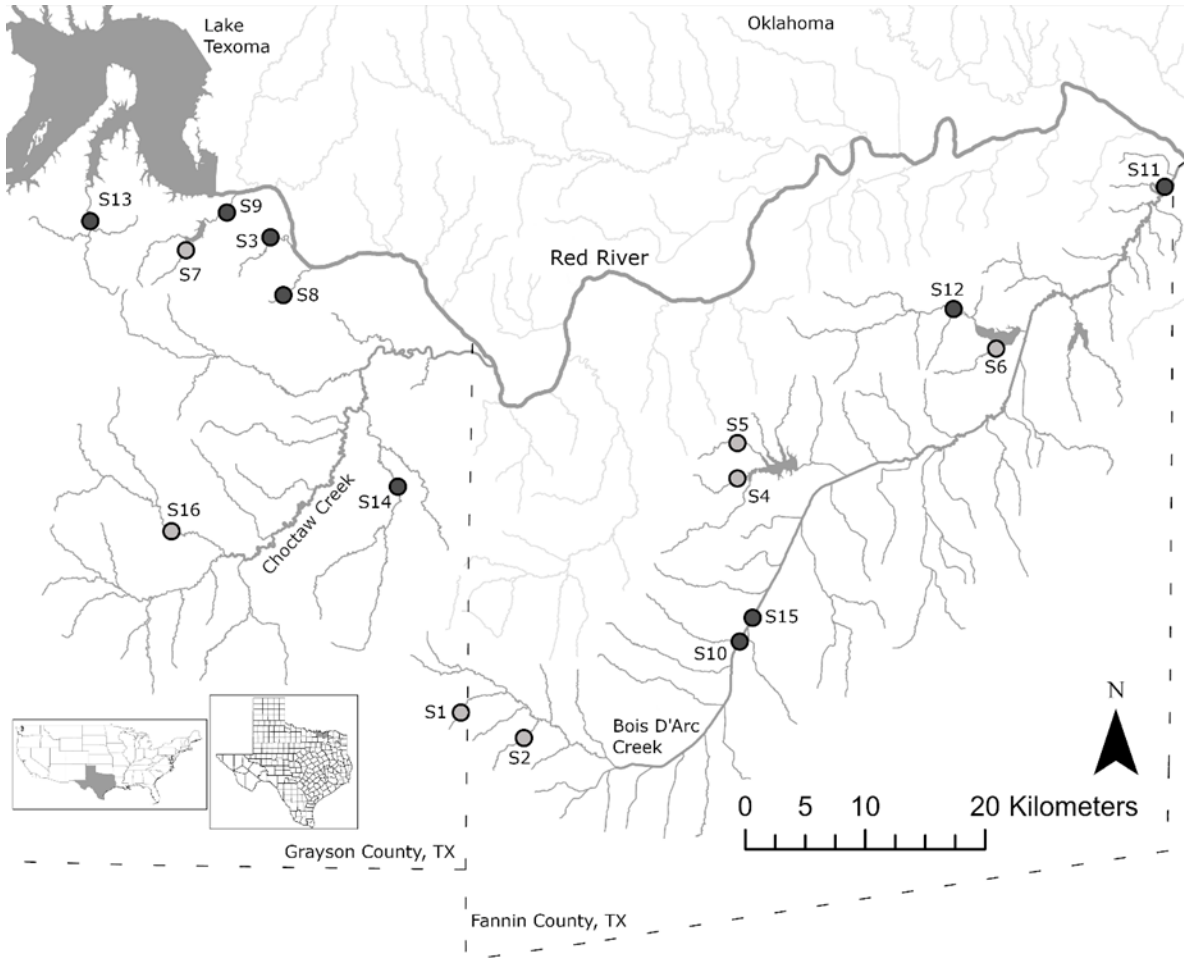


Fig. 1. Map of study sites sampled along middle Red River tributaries in North Texas (Grayson and Fannin Counties). Higher risk sites based on flow patterns are represented in light gray and lower risk sites are represented in black.

Flow Patterns and Drought Risk Groups

For each site, we assessed drought risk based on annual flow patterns from the National Hydrography dataset (NHDplus version 2, USGS). Drought risk was assessed based on annual low flow patterns, and stream flow stability (variance), reflective of stream dependence on groundwater vs rainfall inputs. For low flow values, we used average flow values in August, which had the lowest annual flows in the region. For flow variation, we measured annual Coefficient of Variation in flow values (CV flow) per site to ensure standardization across small and large flow streams. Streams with lower annual low flow values and streams with higher annual flow

variance were placed in a higher drought risk category, as these would indicate a greater likelihood of stream drying during drought. For each variable, we converted the dataset into ranks that indicated drought risk, with higher values in this system indicating higher drought/drying risk. For the low flow values, lower values were given higher risk ranks, and for the CV flow data, higher values were given higher risk ranks (less flow stability through the year). We then took an average of these two scores for each site as drought risk score for each site. For comparative analysis, we split the data into two groups based on a break point in the drought risk scores close to the midpoints of the ranks (break between value of 7 and 8.5). This created a low risk group that all had drought risk scores below 8, which would be the midpoint of 16 sites, ($n = 9$, average risk score = 6.2), and a high risk group that all had drought risk scores above 8 ($n = 7$, average = 11.5) (Table 1).

Community Change

Community change was assessed between paired pre and post drought surveys through the following methods. 1) Sorenson Dissimilarity Index values were used to assess changes in species presence/absence in paired datasets. This method used the following equation: $1 - (2C / (S1 + S2))$, with C equal to the number of species found in both surveys, $S1$ equal to the total number of species in the pre-drought survey, and $S2$ equal to the total number of species in the post-drought survey. 2) Paired change in Shannon Diversity Index values were used to assess changes in diversity based on species richness and evenness. For this metric, a Shannon Diversity value was calculated for both the pre and post drought surveys with the following equation: $H = -\sum p_i * \ln(p_i)$, with p_i representing the relative abundance (proportion of the total catch) for each species. Community change was then calculated as $H_{\text{post}} - H_{\text{pre}}$ for each site to calculate the paired change in Shannon Diversity. 3) Species Gains was calculated as the number of new species that were present at a site in 2023, but absent in 2022, divided by the total number of unique species in both surveys, which standardizes this metric across sites with different species richness values (following implementation in the R package *codyn*). 4) Species Losses was calculated as the number of species that were present at a site in 2022, but absent in 2023, divided by the total number of unique species in both surveys (R package *codyn*). 5) Percent Change in Total Abundance was calculated by summing the abundance across all species at a site, then subtracting the pre-drought total from the post-drought total for each location, then dividing this difference by the total abundance in 2022 to standardize the value across large and small sites with different raw abundance patterns. 6) Paired Euclidian Distance on a Non-metric multidimensional scaling ordination of community patterns was performed. NMDS analysis was done using 32 surveys (pre and post data combined) based on Bray-Curtis dissimilarity indexes across surveys in the dataset. Surveys with similar communities will be placed closer to one another in the ordination. Then, for each site the straight-line Euclidean distance between paired pre and post drought surveys was calculated from the ordination. Sites that had little annual community change would have shorter distances, while sites with greater change would have longer distances in the ordination. We used the statistical software R to conduct two sample t-tests or Wilcoxon Rank-Sum tests (Percent change in Abundance and Paired Euclidean Distance), comparing patterns between the high and low drought risk groups identified by flow patterns.

Life History Trait Analysis

For each species in the dataset, we compiled values for six life history trait variables, relative to opportunistic life history strategies: maximum total length (TL), average spawning season (SS), average age at maturation (MA), maximum longevity (LO), average maximum fecundity (FE), and parental care vs non-parental care after spawning (PC). Values for these traits were compiled from the Fish Traits Database (Frimpong and Angermeier, 2009) and the FishBase database (Froese and Pauly, 2012). If multiple reports were available for fecundity and spawning season, then averages were taken for these values. We then followed the approach of Hitt (2020) and Hitt et al. (2022) to perform an NMDS analysis to ordinate species based on life history trait variables. The NMDS was performed on $\log(x+1)$ transformed trait values, followed by production of a Bray-

Curtis dissimilarity matrix between species, which was used to implement the NMDS ordination (Hitt et al. 2020). Assessment of the resulting ordination showed a triangular pattern, representing a continuum of life history strategies between opportunistic, periodic, and equilibrium strategies. Species that were clustered toward the opportunistic side of the triangle were grouped together as a more opportunistic group, while other species were categorized in a less opportunistic group. Significant differences across these two groups for life history trait values was assessed with Analysis of Similarities (ANOSIM) to ensure that life history patterns were different in the two designated groups.

To compare the annual response of the more opportunistic group vs the less opportunistic group to severe drought, mean relative abundance was calculated for each species in each survey. Relative abundance was used instead of raw abundance to standardize patterns between small and large streams with different total abundance values in the dataset. The average change in relative abundance after drought was then calculated by subtracting the average pre drought relative abundance for a species from the average post drought relative abundance. The average of this value was then taken for each species across all streams where the species was present in either 2022 or 2023 surveys. The envfit function (vegan package, R) was used to assess patterns in mean relative abundance change in relation to life history variables of the species, by adding a vector on the life history trait NMDS plot. A two-sample t-test was also done to compare mean relative abundance change between the more opportunistic group of species and the less opportunistic group of species to assess if the response in abundance after drought was different in these two groups.

RESULTS

Flow Risk Group Analysis

When assessing patterns with Paired Sorenson Dissimilarity Index values, there was a significant difference between high risk (mean = 0.52 ± 0.18) and low risk (mean = 0.31 ± 0.14) groups ($t = -2.17$; $p = 0.05$) (Fig2A). Because this index is based on shared presence/absence of species, this indicated that there was more significant turnover in species following drought in streams with higher risk of drying based on low flow values and flow variance. Breaking down this turnover pattern into species gains and losses, there was not a significant difference in standardized gain values between higher risk (mean = 0.21 ± 0.22) and lower risk (mean = 0.20 ± 0.08) groups ($t = -0.15$; $p = 0.88$). This indicated that the pattern of species gains after drought was not related to flow risk groups, with similar incidence rate across both groups. For standardized species losses, there was a significant difference between higher risk (mean = 0.45 ± 0.20) and lower risk (mean = 0.25 ± 0.12) groups ($t = -2.16$; $p = 0.05$) (Fig2B). This indicated that the difference in species turnover patterns across groups was driven by increased pattern of species losses in higher risk sites. For change in Shannon Diversity values, both sites indicated an average decrease in the diversity metric after drought, and there was no significant difference between higher (mean = -0.26 ± 0.70) and lower risk (mean = -0.23 ± 0.173) groups ($t = 0.1185$; $p = 0.91$), in part because of high variability in this metric across sites. There was no difference in percent change in abundance following drought between high risk (median = -0.10) and low risk (median = -0.28) groups ($W = 22$; $p = 0.351$).

The NMDS analysis of the community patterns across the two survey years, showed relatively high variability across sites, with some sites showing large community changes after drought (highest distance: S1, S2, S4) and others showing less community change (lowest distance: S8, S10, S13) based on abundance and presence/absence patterns (Fig. 3, Panel A). There was a significant difference in Euclidean distance between higher risk (median = 0.57) and lower risk (median = 0.32) site groupings ($W = 10$; $p = 0.02$) (Fig. 2C). This indicates that there was a relationship between flow pattern groupings and community change patterns based on Bray-Curtis Dissimilarity metric, with streams that had lower and more variable annual flows experiencing greater community change.

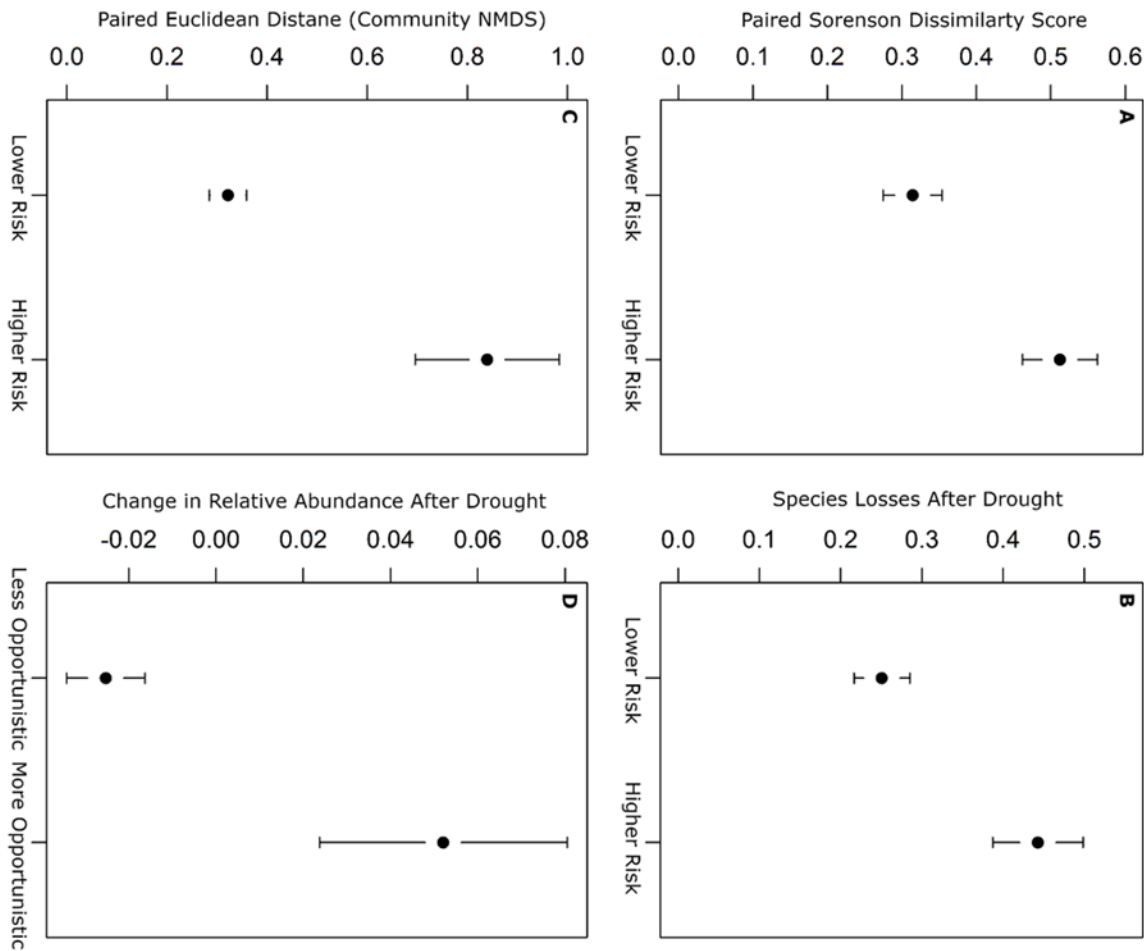


Fig. 2. Comparative analyses across flow patterns or species life history traits (error bars are for SE). A: Community change based on species turnover assessed by Paired Sorensen Dissimilarity scores between lower risk and higher risk groups. B: Patterns of community change based on standardized species losses between lower risk and higher risk groups. C: Patterns of community change based on paired Euclidean distance in combined NMDS analysis of data from both survey years across lower risk and higher risk groups. D: Comparison of patterns of change in mean relative abundance between the more opportunistic and less opportunistic groups in the data.

Life History Trait Analysis

The life history trait analysis with NMDS produced a triangular cluster that represented patterns associated with a more opportunistic (upper right), more periodic (upper left), and a more equilibrium (lower center) strategy (Fig. 3, Panel B). The placement of the six life history variables on the NMDS in relation to life history strategies closely matched previous applications of this method with different communities (Hitt et al. 2022), with opportunistic species in particular being associated with a prolonged spawning season, higher fecundity, decreased longevity, and decreased maximum length. The NMDS plot identified a cluster of 8 species associated with a more opportunistic strategy among the community in the dataset: *Gambusia affinis*, *Notropis buechanani*, *Pimephales vigilax*, *Cyprinella lutrensis*, *Cyprinella venusta*, *Fundulus notatus*, *Etheostoma gracile*, and *Notropis stramineus*. It should be clarified that these patterns represent a continuum, and the patterns observed on the NMDS are specific to the overall variable patterns for these traits that are seen specifically in fish communities of this dataset. ANOSIM indicated that this more opportunistic cluster was significantly different in life history trait values from the less opportunistic group ($R = 0.377$, $p < 0.01$).

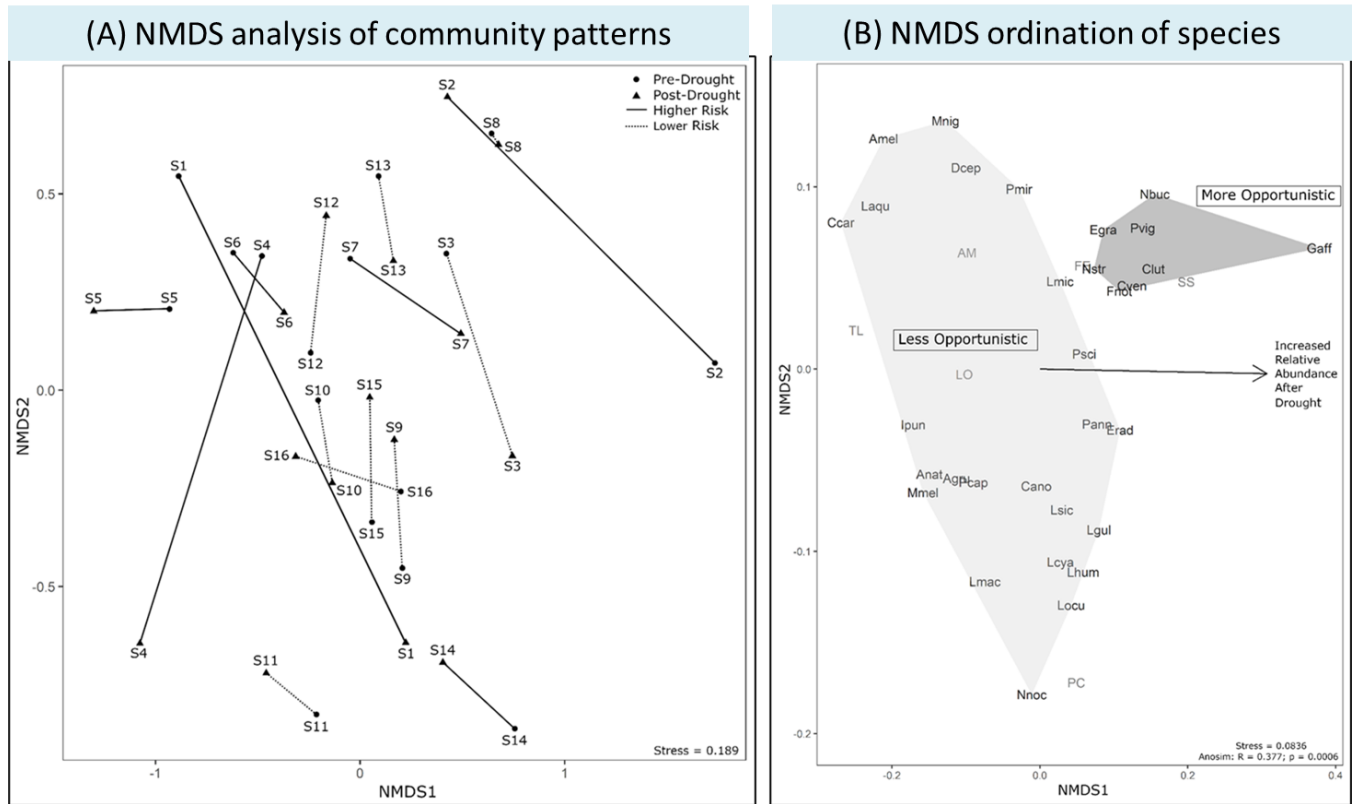


Fig. 3. Panel (A): NMDS analysis of community patterns in two years of surveys (pre drought and post drought). Lines connect surveys for the same location, and Euclidean distance was calculated for the length of the line in NMDS space as an indicator of community change following drought, based on Bray-Curtis dissimilarity matrix across samples. Symbol type designated survey year, and line type designates drought risk group based on flow patterns. Note that the axes have different scales, and therefore distance along NMDS 1 will contribute to a higher Euclidean distance than the same length along NMDS 2. **Panel (B):** NMDS ordination of species based on log-transformed life history trait variables. Variables include: TL: total length, SS: spawning season, MA: maturation age, LO: longevity, FE: fecundity, and PC: parental care. Species codes represent the first letter of the genus and first three letters of the species, with a full community list available in Table 2 for reference. The clusters correspond to species that were more associated with the opportunistic section of the ordination (dark grey) and those associated with less opportunistic traits (light grey), with significant differences across groups determined by ANOSIM. The envfit package (vegan, R) was used to determine which direction of the ordination was associated with increased mean relative abundance following drought as indicated by the vector arrow.

The change in mean relative abundance after drought showed relatively high variability across species (Table 2). The species that increased the most in relative abundance after drought in sites of occurrence were *Notropis bethanani* (ghost shiner), *Cyprinella lutrensis* (red shiner), *Lepomis macrochirus* (bluegill), *Gambusia affinis* (western mosquitofish), and *Fundulus notatus* (blackstripe topminnow). The species who decreased the most in relative abundance after the drought in sites of occurrence were *Etheostoma radiosum* (orangebelly darter), *Labidesthes sicculus* (brook silverside), *Camptostoma anomalum* (central stoneroller), *Pimephales vigilax* (bullhead minnow), and *Cyprinella venusta* (blacktail shiner). Two analyses for change in mean relative abundance following drought suggest that more opportunistic species tended to increase in abundance after drought (mean = 0.066 ± 0.113 ; median = 0.038) and less opportunistic species tended to decrease in abundance (mean = -0.026 ± 0.028 , median = -0.005) groups ($W = 125$; $p = 0.05$) (Fig. 2D). The vector for mean relative abundance on the life history trait NMDS from the envfit function also supports that life history variables associated with more opportunistic strategies were associated with increased abundance after drought, particularly variables more strongly associated with NMDS axis one (Spawning Season, Total Length, Longevity, and Age of Maturation) (Fig. 4, Panel B). For these variables, fishes with shorter maximum total length,

decreased longevity, decreased age of maturation, and prolonged spawning season had more increase in abundance in sites of occurrence after drought. Based on placement of the vector parallel with NMDS axis 1, Fecundity and Parental Care patterns were not as related to change in mean relative abundance as the other variables, as they were placed closer to a value of zero on this axis.

DISCUSSION

In this study, we found that short term community change following severe drought could be related to specific flow variables of sites, with streams that had lower and more variable annual flows having greater change. This pattern was supported by two different measurements of community change (paired Sorenson Index, Paired Euclidean distance on NMDS). Further analysis regarding change in presence/absence of specific species as indicated in the Sorenson Index findings showed that the pattern was mainly driven by species losses following drought, which were higher in sites with lower and more variable flows. Experimental studies have shown a link between water depth, flow, and species losses following simulated drought conditions (Driver and Hoeinghaus, 2016). Likewise, other studies have shown a decline in species richness in monitored sites during drought, relative to depth of refugia or total wetted habitat area that was available during the drought (Hopper et al., 2020; Chang et al., 2024).

Some mechanisms that may relate to this pattern are that streams with more stable, less variable flows on an annual basis may have a greater contribution from groundwater springs. Spring-fed systems can be less affected by fluctuations from rainfall or drought events, which can lead to more stable, less variable flows (Sear et al., 1999). This stability can prevent drying during drought and lead to more stable ecological communities that can resist species change during regional disturbances (Sakai et al., 2021). Although we did not measure comparative influence of groundwater vs runoff for flow patterns in streams, there were several indications in the data that showed the role of groundwater influence on flows at mitigating risk of community changes following drought. For example, Paw Paw Creek (S8) has a strong groundwater influence, being fed by Sand Springs (Baker, 1963). This site had the second lowest flow variability measurement in the data and the fourth smallest annual low flow value. Despite having lower flow, which would make the stream vulnerable to drying, the stability from the spring system caused the community to remain largely unchanged following the drought event (Fig. 3, panel A). This illustrates the importance of groundwater flows at mitigating drought disturbances and the need for incorporating measurements of groundwater inputs or flow variation when assessing drought impacts or risk for fish communities.

The study also showed a difference in short term drought effects on relative abundance of more opportunistic species and less opportunistic species. This result is similar to findings in other studies, and suggest that opportunistic traits can infer drought resistance or resilience on a short-term time scale (Malone et al., 2022). Additionally, the findings suggest that prolonged spawning season and earlier age of maturation were tied to an increase in abundance after the drought in our study. This might suggest that recolonization and rapid reproduction following return of water flows may be driving the community patterns in the study. Anecdotally, many of the communities in the high risk sites had a high proportion of juveniles, which would support this inference, but further testing would be needed to explore if recolonization was the mechanism behind the drought effects seen in the region.

For specific species responses, *N. buchanani*, *C. lutrensis*, *L. macrochirus*, *G. affinis*, and *F. notatus* appeared to be the most resistant to drought effects in our study based on relative abundance changes, and four out of five of these species were grouped in the more opportunistic section of the NMDS plot. *Lepomis macrochirus* was the exception to this, and illustrates that there may be some stochasticity in species responses to disturbances unrelated to life history trait patterns. Additionally, this species is regularly stocked in reservoirs of the region, which might assist with recolonization patterns of streams connected to reservoirs, and they are a highly generalist species that is tolerant of many of the stressors associated with drought refugia (Massure,

2016). *Cyprinella lutrensis* has been shown to have a resistance and recolonization strategy to drought conditions, whereby they may remain in shallow, upstream refugia instead of moving downstream in the network, which can facilitate early colonization of new habitats when waters return (Archdeacon et al., 2022). *Gambusia affinis* has been found to increase greatly in abundance during drought events due to their opportunistic life history traits and their ability to tolerate and even thrive in extreme environmental conditions (Hubbs, 2000, Matthews and Marsh-Matthews, 2011, Hopper et al., 2020). *Notropis buchanani* and *F. notatus* have also previously been described as having opportunistic characteristics that have allowed them to increase in abundance during drought events (Stanley et al., 2012, Schneider, 2019).

Etheostoma radiosum, *L. sicculus*, *C. anomalum*, *P. vigilax*, and *C. venusta* had the lowest decrease in relative abundance in the dataset, and therefore may be the most short-term drought sensitive species in the study. Mollenhauer et al. (2022) performed a long term analysis of species-specific responses to flow patterns in the Red River drainage, and found that *E. radiosum* was negatively associated with flow variability, and *C. anomalum* was negatively associated with disturbance, which both agree with our findings for short term abundance change in drought sensitive sites. *Pimephales.vigilax* has previously been shown to benefit from drought conditions, which would correlate with their placement in the more opportunistic group in this study (Nguyen et al., 2023). The results seen in our study suggest that there is high variance in abundance change patterns across sites for this species, leading to an overall average negative score for mean relative abundance, although there were several sites (3 out of 8 occurrences) where they did have a substantial increase in abundance. Therefore, the response of this species to drought may have several unexplored interactive factors, and appears to be site specific in the study, with an average decrease and high variability across sites. It also important to note that this study examined changes after one year, and communities may respond to drought or other impacts with a loose equilibrium around longer time scales (Matthews), and therefore these species may show longer term resilience relative to their life history traits and dispersal/colonization patterns.

Our research shows that annual low flow and flow variability metrics can be linked to short-term community change or resistance to severe droughts. Streams grouped by these metrics followed predictable differences in drought effects, supported by two different measures of community change. Furthermore, analyses showed that these short term changes were largely driven by presence/absence patterns and species losses, rather than changes in abundance or evenness. The exact mechanism for this pattern was outside the scope of the study and should be explored further, but may be related to rapid colonization and reproduction establishing high abundance in open habitats. The study also showed that within the broader regional community more opportunistic species tended to have a net increase in abundance after drought, while less opportunistic species tended to have a net decrease, with spawning season and age of maturation being important life history traits related to this pattern. These findings can be used to help understand and predict responses to future droughts under different climate scenarios. Current modeling of drought-driven patterns in the Red River through 2050 under difference climate scenarios show high variability across species (Gill et al., 2020). A better understanding of life history based responses to drought and the role of stream stability/resistance to drying on community changes can lead to better precision for predictions and management strategies for drought-prone systems. Future studies should continue to monitor populations in the Red River after drought to determine if communities are maintained around a loose equilibrium or if directional change is occurring, and to successional patterns of recovery in affected communities (Matthews and Marsh-Matthews, 2016).

ACKNOWLEDGEMENTS

This research was funded by internal awards from the College of Science and Engineering at East Texas A&M University. Data collection followed Texas Parks and Wildlife scientific collection permit SPR-1220-211. Protocols for the study were approved by East Texas A&M IACUC review (P22-010)). The authors would like to thank Rahul Rai, Laritza Tinajero, Tina Reichner, and Victoria Hardemann for their assistance in field collections for the study. The authors have no conflicts of interest to declare.

TABLES

Table 1. Flow variables for drought risk classification across sites (data from NHDplus v2). Low Annual flow is in cubic feet per second, and is the average flow in the month of August. CV flow is coefficient of annual monthly variation in flow across one year, with higher values indicating more variability throughout the year, indicating higher influence of precipitation on flow patterns and potentially higher risk of stream drying during low flow conditions.

Site	Group	CV Flow	CV Flow Risk Score	Low Annual Flow (cfs)	Low Flow Risk Score	Avg. Risk Score
S9	lower risk	0.661	1	3.22	8	4.5
S12	lower risk	0.668	6	6.83	6	6
S13	lower risk	0.667	5	3.45	7	6
S16	lower risk	0.675	7	6.97	5	6
S3	lower risk	0.661	3	1.14	10	6.5
S10	lower risk	0.680	10	22.81	3	6.5
S11	lower risk	0.681	12	90.09	1	6.5
S15	lower risk	0.681	11	28.88	2	6.5
S8	lower risk	0.661	2	0.85	12	7
S2	higher risk	0.677	8	1.16	9	8.5
S7	higher risk	0.662	4	0.69	13	8.5
S14	higher risk	0.692	16	7.45	4	10
S1	higher risk	0.679	9	0.43	16	12.5
S5	higher risk	0.682	14	0.89	11	12.5
S6	higher risk	0.681	13	0.65	14	13.5
S4	higher risk	0.684	15	0.65	15	15

Table 2. Mean change in relative abundance (RA) after drought for each species in the total regional dataset, across sites of occurrence in either pre or post drought surveys.

Species (All)	Mean RA change
<i>Etheostoma radiosum</i>	-0.224
<i>Labidesthes sicculus</i>	-0.205
<i>Campostoma anomalum</i>	-0.108
<i>Pimephales vigilax</i>	-0.046
<i>Cyprinella venusta</i>	-0.042
<i>Phenacobius mirabilis</i>	-0.041
<i>Lepisosteus oculatus</i>	-0.028
<i>Lepomis gulosus</i>	-0.025
<i>Ameiurus natalis</i>	-0.022
<i>Lepomis humilis</i>	-0.022
<i>Noturus nocturnis</i>	-0.017
<i>Notropis stramineus</i>	-0.015
<i>Percina sciera</i>	-0.009
<i>Aplodinotus grunniens</i>	-0.005
<i>Ictalurus punctatus</i>	-0.005
<i>Minytrema melanops</i>	-0.005
<i>Cyprinus carpio</i>	-0.003
<i>Dorosoma cepedianum</i>	<0.001
<i>Pomoxis annularis</i>	0.002
<i>Etheostoma gracile</i>	0.002
<i>Micropterus nigricans</i>	0.002
<i>Lepomis aquilensis</i>	0.007
<i>Lepomis cyanellus</i>	0.011
<i>Lepomis microlophus</i>	0.017
<i>Percina caprodes</i>	0.019
<i>Ameiurus melas</i>	0.026
<i>Fundulus notatus</i>	0.039
<i>Gambusia affinis</i>	0.049
<i>Lepomis macrochirus</i>	0.053
<i>Cyprinella lutrensis</i>	0.110
<i>Notropis buechanani</i>	0.320

REFERENCES

- Archdeacon, T.P., Gonzales, E.J., and Thomas, L.I., Movement of Red Shiner during a regulated, intentional surface-flow recession, *Ecol. Freshwater Fish*, 2022, vol. 31, p.358. <https://doi.org/10.1111/eff.12635>
- Baker, E.T., *Groundwater geology of Grayson County, Texas*. 1963, US Department of the Interior Geological Survey, No. 1646.
- Chang, T. Wu, C., Lin, P., and Gao, X. Sudden change in fish functional compositions during a drought in the wet season of the middle Yangtze River, China, *Global Ecol. Conserv.*, 2024, e013144.
- Craig, C.A. and Bonner, T.H., 2019. Drainage basin checklists and dichotomous keys for inland fishes of Texas. *ZooKeys*, 874, p.31.

Dodds, W.K., Gido, K., Whiles, M.R., Fritz, K.M. and Matthews, W.J., Life on the edge: the ecology of Great Plains prairie streams, *BioSci.*, 2004, vol. 54, p.205. [https://doi.org/10.1641/00063568\(2004\)054\[0205:LOTETE\]2.0.CO;2](https://doi.org/10.1641/00063568(2004)054[0205:LOTETE]2.0.CO;2)

Driver, L.J. and Hoeinghaus, D.J. Fish metacommunity responses to experimental drought are determined by habitat heterogeneity and connectivity, *Freshwater Biol.*, 2016, vol. 61, p. 533.

Frimpong, E.A. and Angermeier, P.L., Fish traits: a database of ecological and life-history traits of freshwater fishes of the United States, *Fish.*, 2009, vol. 34, p.487. <https://doi.org/10.1577/1548-8446-34.10.487>

Froese, R. and Pauly, D., 2012. FishBase. World Wide Web electronic publication. www.fishbase.org.

Gill, K.C., Fovargue, R.E., and Neeson, T.M., Hotspots of species loss do not vary across future climate scenarios in a drought-prone river basin, *Ecol. Evol.*, 2020, vol 10, p. 9200.

Hitt, N.P., Landsman, A.P., and Raesly, R.L., Life history strategies of stream fishes linked to predictors of hydrologic stability, *Ecol. Evol.*, 2022, vol. 12, p.e8861. <https://doi.org/10.1002/ece3.8861>

Hitt, N.P., Rogers, K.M., Kelly, Z.A., Henesy, J., and Mullican, J.E., Fish life history trends indicate increasing flow stochasticity in an unregulated river, *Ecosphere*, 2020, vol. 11, p.e03026. <https://doi.org/10.1002/ecs2.3026>

Hodges, S.W., *Movement, survival and refuge use of three minnow species (Campostoma anomalum, Notropis boops, and Semotilus atromaculatus) during seasonal drying in an intermittent Ozark mountain stream*, University of Arkansas: ProQuest, 2007.

Hopper, G.W., Gido, K.B., Pennock, C.A., Hedden, S.C., Frenette, B.D., Barts, N., Hedden, C.K., and Bruckerhoff, L.A., Nowhere to swim: interspecific responses of prairie stream fishes in isolated pools during severe drought, *Aquat. Sci.*, 2020, vol 82, p.1. <https://doi.org/10.1007/s00027-020-0716-2>

Hubbs, C., Survival of *Gambusia affinis* in a Hostile Environment, *Southwest. Nat.*, 2000, vol. 45, p. 521. <https://doi.org/10.2307/3672601>

Hughes, R.M., Wildman, R.C., Gregory, S.V. and Gammon, J.R., Changes in fish assemblage structure in the main-stem Willamette River, Oregon, *Am. Fish. Soc. Symp.*, 2005, vol. 45, p. 61.

Jones, N.E. and Petreman, I.C., Relating extremes of flow and air temperature to stream fish communities, *Ecohydrol.*, 2013, vol. 6, p.826. <https://doi.org/10.1002/eco.1305>

Keaton, M., Haney, D., and Andersen, C.B., Impact of drought upon fish assemblage structure in two South Carolina Piedmont streams, *Hydrobiol.*, 2005, vol. 545, p. 209. <https://doi.org/10.1007/s10750-005-2674-z>

Lennox, R.J., Crook, D.A., Moyle, P.B., Struthers, D.P., and Cooke, S.J., Toward a better understanding of freshwater fish responses to an increasingly drought-stricken world, *Rev. in fish bio. and fish.*, 2019, vol. 29, p.71. <https://doi.org/10.1007/s11160-018-09545-9>

Magoulick, D.D., Dekar, M.P., Hodges, S.W., Scott, M.K., Rabalais, M.R., and Bare, C.M., Hydrologic variation influences stream fish assemblage dynamics through flow regime and drought, *Sci. rep.*, 2021, vol. 11, p.10704. <https://doi.org/10.1038/s41598-021-89632-3>

Magoulick, D.D. and Kobza, R.M., The role of refugia for fishes during drought: a review and synthesis, *Freshw. Bio.*, 2003, vol. 48, p.1186. <https://doi.org/10.1046/j.1365-2427.2003.01089.x>

Malone, E.W., Perkin, J.S., Gibbs, W.K., Padgett, M., Kulp, M., and Moore, S.E. High and dry in days gone by: Life-history predicts Appalachian mountain stream fish assemblage transformation during historical drought, *Ecol. Freshwater Fish*, 2022, vol 31, p. 29.

Marshall, J.C., Lobegeiger, J.S., and Starkey, A., Risks to fish populations in dryland rivers from the combined threats of drought and instream barriers, *Front. Environ. Sci.*, 2021, vol. 9, p. 671556. <https://doi.org/10.3389/fenvs.2021.671556>

Massure, W.A., Evaluating effects of drought and anthropogenic influences on the growth of stream fishes on the Edwards Plateau, Central Texas. Ph. D. Thesis, Texas Tech University, 2016.

Matthews, W.J. and Marsh-Matthews, E., Effects of drought on fish across axes of space, time and ecological complexity, *Freshw. Bio.*, 2003, vol. 48, p.1232. <https://doi.org/10.1046/j.1365-2427.2003.01087.x>

- Matthews, W.J. and Marsh-Matthews, E. An invasive fish species within its native range: community effects and population dynamics of *Gambusia affinis* in the central United States. *Freshwater Biol.*, 2011, vol. 56, p.2609. <https://doi.org/10.1111/j.1365-2427.2011.02691.x>
- Matthews, W.J. and Marsh-Matthews, E., Dynamics of an upland stream fish community over 40 years: trajectories and support for the loose equilibrium concept, *Ecol.*, 2016, vol. 97, p.706. <https://doi.org/10.1890/14-2179.1>
- Mollenhauer, R., Mouser, J.B., Roland, V.L., and Brewer, S.K. Increased landscape disturbance and streamflow variability threaten fish biodiversity in the Red River catchment, USA, *Divers. Distrib.*, 2022, vol. 28, p. 1934.
- National Oceanic and Atmospheric Administration (NOAA), 2022. National Integrated Drought Information System. NOAA. Retrieved from <https://www.drought.gov/states/Texas>
- Nguyen, E., Mayes, K.B., Smith, R., Trungale, J., and Perkin, J.S., The duality of drought: Pelagic-and benthic-spawning stream fishes show opposing responses to drought in the southern Great Plains, *North Am. J. Fish. Manage.*, 2023, vol. 43, p.1276. <https://doi.org/10.1002/nafm.10874>
- Perkin, J.S., Gido, K.B., Costigan, K.H., Daniels, M.D. and Johnson, E.R. Fragmentation and drying ratchet down Great Plains stream fish diversity. *Aquat. Conserv. Mar. Freshwater Ecosyst.*, 2015, vol. 25, p. 639.
- Sakai, M., Iwabuchi, K., and Bauman, D. Unique habitat and macroinvertebrate assemblage structures in spring-fed stream: a comparison among clastic lowland tributaries and mainstreams in northern Japan, *Community Ecol.*, 2021, vol. 22, p. 193.
- Schneider, S.J., Seasonal fish assemblage structure in riffles, wadeable pools and non-wadeable pools of a perennial warmwater river, MS Thesis, Emporia State University, 2019.
- Sherwood, S. and Fu, Q., A drier future?, *Sci.*, 2014, vol. 343, p. 737.
- Smith, J.J., Fishes of the Pajaro River system, in Moyle, P.B., J.J. Smith, R.A. Daniels, T.L. Price, and D.M. Baltz, eds., Distribution and ecology of stream fishes of the Sacramento-San Joaquin drainage system, California. Berkeley, CA: University of California Press, 1982, p. 83.
- Stanley, C.E., Taylor, J.M., and King, R.S., Coupling fish community structure with instream flow and habitat connectivity between two hydrologically extreme years, *Trans. Am. Fish. Soc.*, 2012, vol. 141, p.1000. <https://doi.org/10.1080/00028487.2012.675893>
- Texas Commission on Environmental Quality (TCEQ) Water Quality Planning Division, 2014. Surface water quality monitoring procedures, volume 2: Methods for collecting and analyzing biological assemblage and habitat data. *Texas Commission on Environmental Quality*.
- Vicente-Serrano, S.M., Peña-Angulo, D., Beguería, S., Domínguez-Castro, F., Tomás-Burguera, M., Noguera, I., Gimeno-Sotelo, L., and El Kenawy, A., Global drought trends and future projections, *Phil. Trans. R. Soc. A*, 2022, vol 380, p. 20210285. <https://doi.org/10.1098/rsta.2021.0285>
- Wheeler, K., Wenger, S.J., and Freeman, M.C. States and rates: Complementary approaches to developing flow-ecology relationships. *Freshwater Biol.*, 2018, vol. 63, p. 906.
- Winemiller, K.O. and Rose, K.A. Patterns of life-history diversification in North American fishes: implications for population regulation. *Can. J. Fish. Aquat. Sci.*, 1992, vol. 49, p. 2196.